

Water Reminder Notification Strategy

Technical Analysis — iOS & Android Platform Stability

Date: May 19, 2026**Platform:** Flutter (iOS & Android)**Prepared by:** Engineering Team**Version:** 1.0

1. EXECUTIVE SUMMARY

The Triva app delivers water reminder notifications to users at configurable intervals (30 min, 1 hr, 1.5 hrs, 2 hrs) between a user-defined wake-up time and bed time. This document evaluates two implementation approaches — the current WorkManager-based one-shot scheduling and the recommended daily-repeating notification approach — and provides a stability comparison across both iOS and Android platforms.

2. HOW THE FEATURE WORKS

The user configures three settings in the Reminder screen:

Setting	Description	Example
Wake-up Time	Time from which reminders begin each day	7:00 AM
Bed Time	Time at which reminders stop each day	10:00 PM
Reminder Interval	How frequently the reminder fires between the two times above	Every 1 hour

The app then schedules local device notifications that appear even when the app is closed, reminding the user to drink water.

3. APPROACH COMPARISON

Approach A — Current Implementation (One-Shot + WorkManager)

Notifications are scheduled as individual one-time events for each reminder slot of the day. A background task (WorkManager) runs daily to refill the next day's slots. The app also refills on every resume.

Approach B — Recommended (Daily Repeating Notifications)

Each reminder time slot is scheduled once as a **daily-repeating notification** using the OS notification scheduler. The OS automatically fires it every day at the same time — no background task needed. Settings are re-applied only when the user changes them.

Feature / Scenario	Approach A — Current	Approach B — Recommended
iOS Reliability	⚠️ Unreliable — depends on OS allowing background task	✅ Very Stable — OS-native, no background execution
Android Reliability	✅ Stable — WorkManager is Android-native	✅ Stable — Same underlying alarm manager
After Phone Reboot	❌ Lost — reminders stop until user opens app	✅ Restored — OS reschedules automatically
User Does Not Open App for 3+ Days	❌ Stops — 2-day lookahead exhausted	✅ Continues — daily repeat never expires
OEM Battery Optimization (Xiaomi, Huawei, Samsung)	⚠️ Same risk	⚠️ Same risk

Background Execution Required	⚠️ Yes — WorkManager must fire daily	✅ No — zero background tasks for water
iOS Notification Slot Usage (64 limit)	Up to 56 slots per 2-day window (at 30 min)	Up to 28 slots total (daily-repeating, reused)
Code Complexity	High — parity ID system, WorkManager callback, reschedule logic	Low — schedule once on save, cancel on disable
Handles Settings Changes	Via daily reschedule cycle	Cancel all + re-schedule immediately on save
Industry Standard for This Use Case	⚠️ Partial	✅ Yes — used by hydration & habit apps

4. PLATFORM-BY-PLATFORM STABILITY

 Android Overall: Stable on both approaches				
WorkManager support				Native ✅
Daily repeating notifications				Fully supported ✅
Exact alarms (SCHEDULE_EXACT_ALARM)				Supported ✅
Reboot recovery				Yes (Approach B) ✅
OEM battery kill (Xiaomi, Huawei)				Risk on both ⚠️
Recommendation				Approach B preferred
 iOS Approach A: Unreliable Approach B: Very Stable				
WorkManager (BGTaskScheduler)				Unreliable ❌
Daily repeating notifications				Apple-recommended ✅
UNUserNotificationCenter				OS-native ✅
Reboot recovery				Yes (Approach B) ✅
Max local notifications				64 total (iOS limit)
Recommendation				Approach B — only viable option

5. IOS NOTIFICATION SLOT USAGE

iOS limits each app to **64 scheduled local notifications** at any time. The table below shows how many daily-repeating slots are needed per interval setting, and how many slots remain available for other features (medicine & supplement reminders).

Reminder Interval	Waking Hours (example)	Water Slots Used	Slots Remaining (for other features)	Status
Every 30 minutes	7:00 AM - 10:00 PM (15 hrs)	30 slots	34 remaining	✅ Within limit
Every 1 hour	7:00 AM - 10:00 PM (15 hrs)	15 slots	49 remaining	✅ Comfortable
Every 1.5 hours	7:00 AM - 10:00 PM (15 hrs)	10 slots	54 remaining	✅ Comfortable
Every 2 hours	7:00 AM - 10:00 PM (15 hrs)	7 slots	57 remaining	✅ Comfortable

6. KNOWN RISKS & MITIGATIONS

Risk	Affected Platform	Severity	Mitigation
User disables notifications in device Settings	iOS & Android	High	In-app prompt guiding user to re-enable permissions
OEM battery optimization kills alarm delivery (Xiaomi, Huawei, OnePlus)	Android	Medium	In-app guide asking user to whitelist app in battery settings
WorkManager not firing on iOS (BGTaskScheduler unreliable)	iOS only	High (Approach A)	Switch to daily-repeating notifications (Approach B) — eliminates risk
Reminders lost after phone reboot	Android (Approach A only)	Medium	Daily-repeating notifications (Approach B) survive reboots automatically
Exact alarm permission revoked (Android 12+)	Android	Medium	App already requests SCHEDULE_EXACT_ALARM at setup; handle graceful degradation

7. RECOMMENDATION

Adopt Approach B — Daily Repeating Notifications on Both Platforms

- iOS:** Replace WorkManager-based scheduling with daily-repeating local notifications via `UNUserNotificationCenter`. This is the Apple-recommended pattern and eliminates all iOS background execution dependency.
- Android:** Use the same daily-repeating approach. WorkManager can be retained as a lightweight safety net for medicine & supplement reminders only.
- On settings save:** Cancel all existing water notification slots and re-schedule with the new settings. No background task required for ongoing delivery.
- Result:** Reminders fire reliably every day on both platforms, survive phone reboots, and work even if the user hasn't opened the app for weeks.

8. GLOSSARY

Term	Meaning
WorkManager	Android background task library. On iOS it uses BGTaskScheduler, which Apple controls and is not guaranteed to run on schedule.
Local Notification	A notification scheduled entirely on-device. No internet connection or server is required.
Daily Repeating Notification	A local notification configured to automatically fire every day at the same time, managed natively by the OS.
BGTaskScheduler	Apple's system for running background tasks on iOS. It is intentionally limited and non-deterministic to preserve battery life.
UNUserNotificationCenter	Apple's official API for scheduling and managing local & push notifications on iOS.
SCHEDULE_EXACT_ALARM	Android 12+ permission required to schedule notifications at a precise time (as opposed to an approximate window).
Doze Mode	Android power-saving feature that restricts background activity when the phone is idle. Exact alarms are exempt.
OEM Battery Optimization	Manufacturer-specific (Xiaomi, Huawei, Samsung) battery management that can kill background processes more aggressively than stock Android.

